

Sangam Pratap Singh

Gmail | Portfolio | LinkedIn | GitHub | GitLab (KDE) | Codeforces (1326) | +91-9205219678

EDUCATION

ABV-Indian Institute of Information Technology and Management (IIITM), Gwalior

Aug 2023 – May 2028

Integrated Dual Degree - B.Tech in Information Technology + MBA

Gwalior, India

EXPERIENCE

Open-Source Contributor - KDE Plasma

Dec 2024 – Present

KDE | C++, Qt/QML, KConfig

Remote

- **Keyboard Accessibility (MR 6103)**: Enabled “Enter” key navigation on calendar header elements in Plasma Digital Clock, bringing the widget into compliance with WCAG keyboard interaction patterns.
- **Wallpaper Path Persistence (MR 6208)**: Eliminated repeated directory browsing for users by persisting wallpaper picker selections via KConfig; handled edge cases for unmounted drives and moved or deleted folders, reducing friction in a workflow used by millions of KDE desktops.
- **Wallpaper Gallery Caching (MR 6280)**: Introduced local caching with **SHA-256** content-hash deduplication, ensuring wallpapers remain available even when source files are moved or deleted.
- **Todo Checkbox Bug Fix (MR 172: Marknote)**: Resolved a state-inheritance bug where pressing Enter inside a checked list item created a pre-checked child; new items now always initialise as unchecked.
- **Spacebar Pan Shortcut (MR 430: Drawy)**: Implemented press-and-hold spacebar to temporarily activate the move tool for canvas panning, matching the UX convention of tools like Excalidraw.

Software Development Intern

Jul 2025 – Sep 2025

OpenSphere | React.js, Node.js, AWS SES, MongoDB, WebSockets

Remote

- Designed and shipped a **real-time commenting system** inside the platform’s in-house text editor, enabling threaded discussions for concurrent collaborators; editor adoption jumped **5x** post-launch, reflected by API key consumption rising from a **2-month** cycle to **1.5 weeks**.
- Eliminated a data-filing redundancy in the **end-to-end drafting workflow**, providing **2x throughput** for legal operations teams and cutting manual intervention to zero for that stage.
- Streamlined **task-creation pipelines** to improve the document collection pipeline, cut configuration time and reduced client confusion, directly improving the customer experience.
- Expanded platform coverage by implementing **self-petitioning visa workflows**, unlocking a previously unsupported application category and opening access to new client segments.
- Delivered a **user activity tracking dashboard** with granular action logs across the platform, improving operational transparency and enabling compliance audit trails.

PROJECTS

CodeRivals | GitHub | React.js, SpacetimeDB, AWS EC2, Tailwind CSS, Monaco Editor

Apr 2026

- Developed a real-time **1v1 competitive coding platform** where players duel in live matches and strategically deploy **power cards** to influence round outcomes.
- Implemented an **ELO rating system** modelling win probability via logistic function over rating differentials, updating scores after each match to reflect true skill and improve matchmaking quality.
- Deployed a self-hosted **Piston** code execution API on **AWS EC2**, containerizing execution workers, configuring security groups, and exposing judge ports: after abandoning **Judge0** due to **cgroup v2** incompatibility.
- Built a custom **C++ input parser pipeline** to normalize **stdin**/test-case ingestion across varied problem formats, handling edge-case whitespace, multiline inputs, and type coercion to improve judged submission reliability end-to-end.

DevTrackr | GitHub | Go, PostgreSQL, React.js, Resend

Apr 2026

- Built a Go-based **multi-platform activity aggregator** that pulls daily coding data from GitHub, LeetCode, and Codeforces into a unified PostgreSQL schema; **evaluated by 30 real users**, with multiple users reporting measurable improvement in daily coding consistency within 3–4 days of use
- Designed a scheduled pipeline with configurable delivery intervals, generating and dispatching personalized HTML email digests summarizing per-platform contribution metrics.

Astra | GitHub | Electron, React.js, Node.js, Express, MongoDB, Tree-sitter, Gemini CLI

Aug 2025 – Sep 2025

- Engineered a cross-platform **Electron** desktop tool that scans multi-repository codebases and surfaces dependency vulnerabilities at scale via the **OSV.dev API**, filtering out non-essential directories and honouring **.gitignore**; tested against codebases with **10,000+ files**.
- Combined **Tree-sitter AST parsing** with **Gemini CLI** to perform AI-driven static analysis, automatically generating context-aware remediation patches for identified CVEs.
- Automated patch delivery by programmatically creating **feature branches from main/master**, reducing the manual steps in the vulnerability-resolution cycle from identification to pull request.

TECHNICAL SKILLS

- **Languages**: C++, C, Go, JavaScript, Python, Qt/QML
- **Backend & Frameworks**: Node.js, Express.js, React.js, Socket.io, Electron, WebSockets
- **Databases**: PostgreSQL, MongoDB, MySQL, SpacetimeDB
- **Cloud & DevOps**: AWS EC2, AWS SES, Docker, Nginx
- **Tools**: Git, Linux, GitHub Actions